

SHOEBURYNESSE LANDWICK, United Kingdom

WMO: 036930

Lat: 51.5547N

Lon: 0.8269E

Elev: 3

StdP: 101.29

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-2.4	-1.2	-5.2	2.4	-0.2	-3.9	2.7	0.6	13.9	8.8	12.5	7.9	3.6	310	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.9	25.2	18.7	23.7	18.0	22.5	17.3	19.8	23.8	19.0	22.4	18.1	21.4	4.8	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.2	21.7	17.5	12.6	20.9	16.7	11.9	20.1	56.6	23.8	53.8	22.4	51.2	21.5	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.4	10.2	9.1	DB	-4.9	29.2	1.7	1.5	-6.1	30.3	-7.1	31.2	-8.1	32.0	-9.4	33.1	
			WB	-5.2	21.7	1.7	0.9	-6.4	22.4	-7.4	22.9	-8.3	23.5	-9.5	24.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.9	5.1	5.3	6.7	9.1	12.2	15.2	17.7	17.8	15.2	12.2	8.3	5.8
	DBStd	5.37	3.17	3.09	2.84	2.90	2.72	2.51	2.28	2.22	2.37	2.76	3.10	3.41
	HDD10.0	664	152	132	106	49	11	1	0	0	0	11	68	134
	HDD18.3	2751	409	363	360	277	191	98	39	37	96	191	302	388
	CDD10.0	1002	2	2	4	23	79	156	239	241	157	78	16	4
	CDD18.3	47	0	0	0	0	0	4	20	20	3	0	0	0
	CDH23.3	174	0	0	0	0	2	17	76	69	9	2	0	0
	CDH26.7	18	0	0	0	0	0	1	8	8	0	0	0	0
Wind	WSAvg	4.7	5.3	5.2	4.9	4.5	4.5	4.3	4.1	4.2	4.5	4.8	4.8	5.0
Precipitation	PrecAvg	504	39	29	36	36	35	48	40	47	47	50	52	45
	PrecMax	659	121	62	85	66	67	110	103	108	165	130	101	95
	PrecMin	331	11	6	3	4	1	5	8	10	3	2	11	13
	PrecStd	74	21	15	22	17	17	28	23	23	37	33	21	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.0	13.7	15.3	20.0	22.4	25.5	27.7	27.7	24.4	20.4	16.1	13.9
		MCWB	11.3	10.9	11.1	14.6	16.7	19.0	19.3	20.0	18.6	16.4	14.3	12.3
	2%	DB	11.9	12.0	13.5	17.1	19.9	23.1	25.2	25.1	22.2	18.3	14.7	12.8
		MCWB	10.2	9.9	10.2	12.6	15.3	17.4	18.7	18.8	17.5	15.4	12.9	11.2
	5%	DB	10.8	10.9	12.3	15.2	18.4	21.4	23.7	23.5	20.6	17.1	13.8	11.9
		MCWB	9.4	8.9	9.5	11.4	14.2	16.5	18.0	18.0	16.3	14.6	12.0	10.5
	10%	DB	9.8	9.9	11.3	13.9	17.1	19.9	22.5	22.1	19.3	16.2	12.7	10.9
		MCWB	8.4	8.1	8.9	10.6	13.3	15.6	17.3	17.1	15.5	14.0	11.2	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.3	11.9	15.1	17.6	20.0	20.9	20.9	19.8	17.1	14.7	12.5
		MCDB	12.6	13.0	14.1	19.3	21.1	24.4	25.7	25.9	22.8	19.5	15.8	13.5
	2%	WB	10.5	10.3	10.9	13.3	16.0	18.4	19.8	19.8	18.3	16.1	13.4	11.6
		MCDB	11.6	11.7	12.8	16.1	19.1	21.8	23.8	23.7	21.1	17.7	14.4	12.6
	5%	WB	9.6	9.3	10.1	12.1	14.9	17.3	18.9	18.9	17.3	15.3	12.4	10.7
		MCDB	10.7	10.6	11.8	14.4	17.6	20.3	22.6	22.2	19.7	16.7	13.5	11.8
	10%	WB	8.6	8.4	9.1	11.1	13.9	16.3	18.0	18.1	16.3	14.5	11.4	9.7
		MCDB	9.7	9.7	10.9	13.4	16.4	19.1	21.5	21.1	18.5	15.8	12.5	10.8
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.3	6.3	7.5	7.6	7.8	8.1	7.9	7.4	6.1	5.2	4.5
		MCDBR	5.2	6.1	7.7	9.2	9.7	9.8	10.1	9.7	8.6	6.6	5.0	4.4
		MCWBR	5.0	4.9	5.3	5.5	5.7	5.5	4.9	4.7	4.8	4.6	4.3	4.1
	5% WB	MCDBR	5.2	5.7	7.1	8.9	9.2	9.1	9.3	8.9	7.9	6.1	4.9	4.6
		MCWBR	5.1	4.9	5.4	5.8	5.9	5.6	5.1	4.8	5.0	4.7	4.5	4.5
Clear-Sky Solar Irradiance	taub	0.333	0.350	0.387	0.408	0.414	0.415	0.413	0.404	0.383	0.366	0.339	0.324	
	taud	2.374	2.377	2.267	2.240	2.268	2.289	2.307	2.353	2.402	2.425	2.416	2.365	
	Ebn at Noon	670	767	799	828	841	841	837	825	800	738	659	613	
	Edh at Noon	63	81	108	125	127	126	121	110	94	77	60	54	
All-Sky Solar Radiation	RadAvg	0.78	1.45	2.57	4.12	4.91	5.44	5.16	4.30	3.23	1.85	0.94	0.60	
	RadStd	0.07	0.22	0.32	0.39	0.35	0.42	0.46	0.33	0.26	0.18	0.11	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page